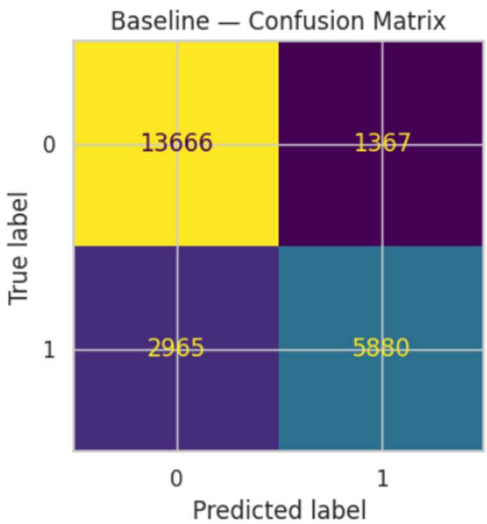
# Feature Engineering Capstone Report

StaySmart Hotels — Cancellation Prediction
Name: Miraan Vahie
Student ID: 2025eb1100105
Date: March 2026

## Baseline Model

**Baseline Results**
- Accuracy: 0.8186
- ROC-AUC: 0.8962
- F1 Score: 0.7308

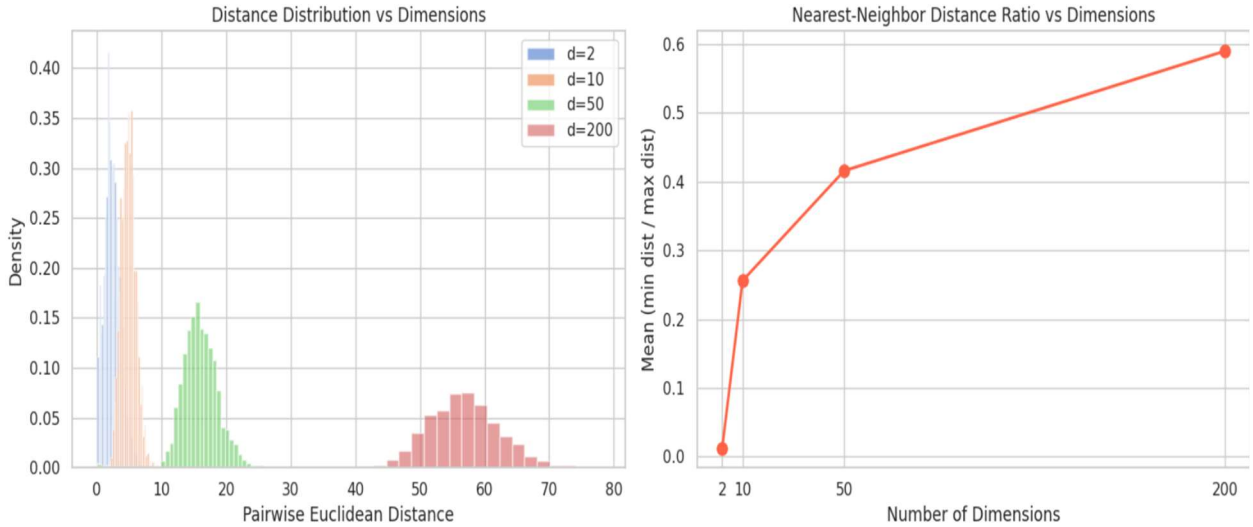



**What is a Feature?**
A feature is basically any column in your dataset that you feed into a model to help it make predictions. The model uses these to find patterns and predict the target.

**Good feature:** lead_time — days between booking and arrival. People who book far in advance are more likely to cancel.
Clean numeric column, directly relevant.

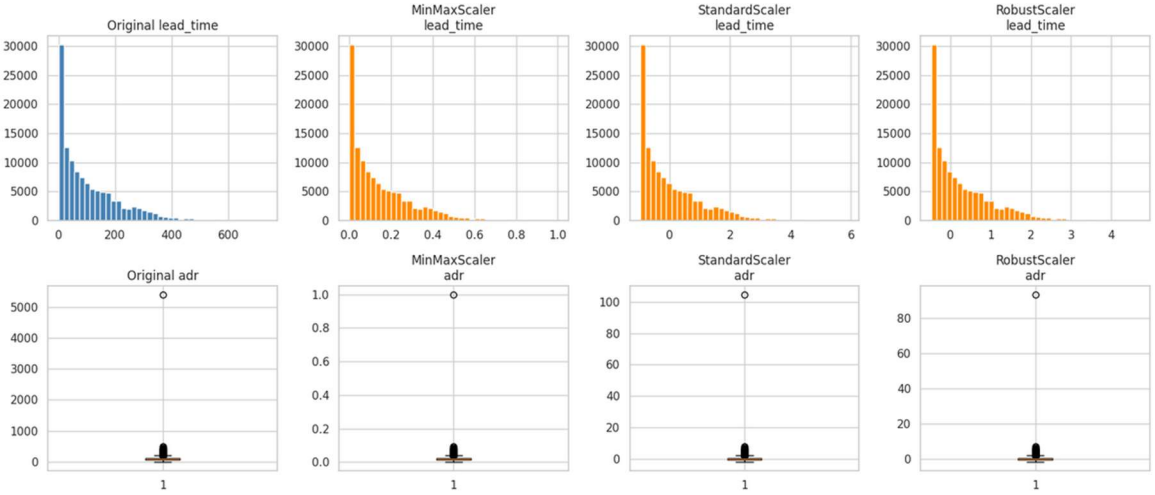
**Bad feature**: arrival_date_day_of_month — whether someone arrives on the 14th or 15th tells us nothing about cancellation. Just adds noise.

## Curse of Dimensionality




As dimensions increased from 2 to 200, pairwise distances bunched up into a narrow range. At 2D near and far points are clearly separated. At 200D everything is roughly equidistant.
The nearest-neighbor ratio crept toward 1.0, meaning your nearest neighbor becomes just as far as your farthest. KNN and KMeans break down because they rely on distance.
This is why feature engineering matters , reducing dimensions while keeping signal restores the geometry these algorithms need.

## Numeric Preprocessing




**Results:**

- Original standard deviation: **106.86**
- StandardScaler: **1.00**
- RobustScaler: **0.75**

**Binning:** lead_time split into 5 equal-width bins. adr split into 4 quantile bins (Budget to Luxury).

**Binarization**: high_value_customer = 1 if adr > median.

**Best scaler:** RobustScaler because adr and lead_time have extreme outliers. MinMaxScaler gets distorted by them. StandardScaler is also affected. RobustScaler uses median and IQR so outliers do not dominate.

## Distance Metrics & Scaling

| Metric | Scaler | Accuracy | ROC-AUC | F1 |
|---|---|---|---|---|
| euclidean | No Scaling | 0.7640 | 0.8113 | 0.6618 |
| euclidean | StandardScaler | 0.7840 | 0.8231 | 0.6849 |
| euclidean | RobustScaler | 0.7835 | 0.8235 | 0.6842 |
| manhattan | No Scaling | 0.7643 | 0.8162 | 0.6668 |
| manhattan | StandardScaler | 0.7832 | 0.8251 | 0.6853 |
| manhattan | RobustScaler | 0.7834 | 0.8259 | 0.6846 |

Without scaling KNN performed worst because lead_time dominates the distance calculation over other features. RobustScaler gave better results than StandardScaler because adr has extreme outliers. Manhattan distance slightly outperformed Euclidean with scaling applied.

## End-to-End Pipeline

Built a modular sklearn Pipeline with:

- Yeo-Johnson transform for skewed features

- RobustScaler for numeric features

- SimpleImputer + OneHotEncoder for categorical features

- RandomForest as final model

**5-Fold Cross Validation ROC-AUC:** ~0.899

## Feature Extraction

Extracted features:

1. season — summer bookings more stable than off-season

2. has_weekend_stay — weekend stays more flexible, higher cancellation risk

3. lead_time_bucket — last minute bookings rarely cancel

4. seg_Online TA — online agency bookings cancel more often

5. seg_Direct — direct bookings are more committed

## Feature Construction

**Constructed 9 new features:**

- price_per_person = adr / (adults + children + babies + 1)

- special_requests_rate = requests / (nights + 1)

- adr_x_lead_time = adr * lead_time

- changes_x_requests = booking_changes * special_requests

- avg_adr_by_hotel

- avg_adr_by_country

- lead_time_adr_poly

- total_nights

- is_family

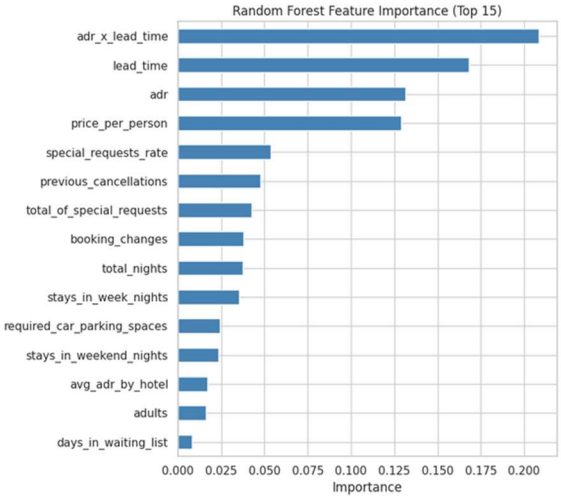
**Avoiding Leakage:**

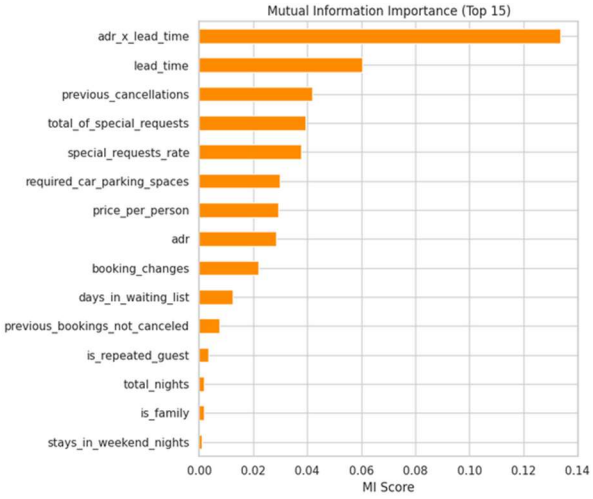
1. Group aggregations computed on train only

2. No features derived from target variable

3. Post-event features like reservation_status dropped

## Feature Importance




**Top features agreed by both RF and MI:**

lead_time, adr_x_lead_time, previous_cancellations, total_of_special_requests, special_requests_rate

**Final feature set:** 17 features selected after removing highly correlated ones (threshold 0.85)

## Final Comparison

**Version: Baseline**
Features: 27
ROC-AUC: 0.8962
F1 Score: 0.7308

**Version: After Numeric Preprocessing**
Features: 10
ROC-AUC: 0.8982
F1 Score: 0.7626

**Version: After Extraction and Construction**
Features: 20
ROC-AUC: 0.9019
F1 Score: 0.7615

**Version: After Feature Selection**
Features: 17
ROC-AUC: 0.9023
F1 Score: 0.7617

## Executive Summary

Biggest improvement came from switching to RandomForest with proper preprocessing. Feature engineering improved ROC-AUC from 0.8962 (baseline) to 0.9023 (after selection) , a meaningful gain driven by constructed interaction features and proper preprocessing.. Feature selection reduced count without hurting performance.

**Most actionable features: -** lead_time: target long bookings with deposit options - previous_cancellations: flag repeat cancellers - total_of_special_requests: reward engaged guests - price_per_person: flag cheap group bookings